

Cover Letter

Hi NestAI team,

I'm a GIS data specialist and PhD candidate at Aalto University, based in Espoo. I use the location database for my research, professional work, and side projects. Over the past few years, I've built end-to-end geospatial data pipelines, ranging from ingesting and correcting building geometry, land use policies, road networks, and SoftGIS data across Finland and the Netherlands for EU-wide deployment to designing DuckDB- and PostGIS-backed data lakes for research and production use. The tools in your stack (GeoParquet, Shapely, GeoPandas, PostGIS, cloud pipelines on GCP and Azure) are things I work with every day. I am a quick study and acquire skills when needed, but I also have a strong foundation in the core tools and concepts of geospatial data management and analysis. I have experience working with confidential data, anonymisation processes, GDPR compliance, and cybersecurity. I have signed way too many NDAs!

What draws me to NestAI specifically is the emphasis on reliability, sovereignty, and self-reliance. My work has always been in environments where the data has to be right as demanded by the research community. For me, spatial integrity is essential, especially when it directly impacts policy decisions. I've also led performance work (an 86% improvement in targeted pipeline calculations through a Pandas-to-Polars migration) and built automated, API-driven workflows that reduce manual intervention.

I am located in Finland and have been living here for eight years. There's no relocation complexity, nor are residence permit requirements on my end. I'd love to chat more about how my background fits what you're working on at NestAI.

Terveisin,

Babäk Firoozi Fooladi